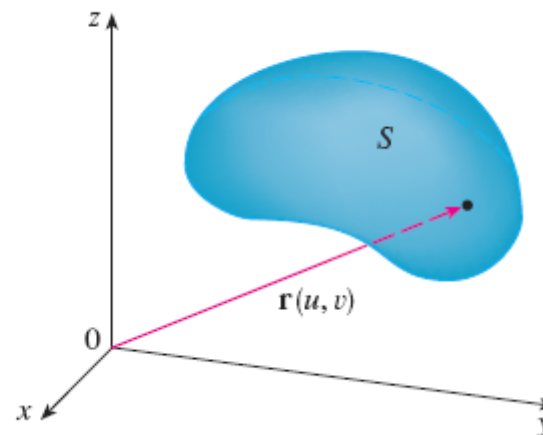
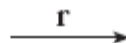
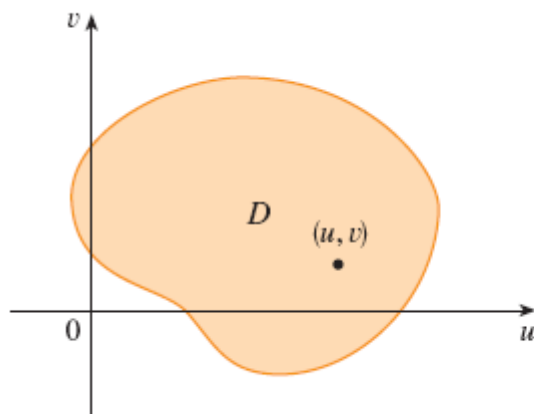
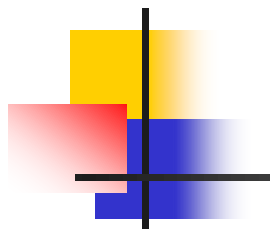


Parametric Surfaces & Their Areas

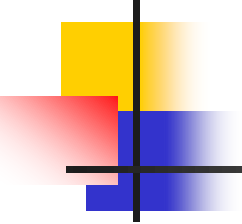


Example:

Determine the surface given by the parametric representation

$$\mathbf{r}(u, v) = u\mathbf{i} + u \cos v\mathbf{j} + u \sin v\mathbf{k}.$$

Parametric Surfaces



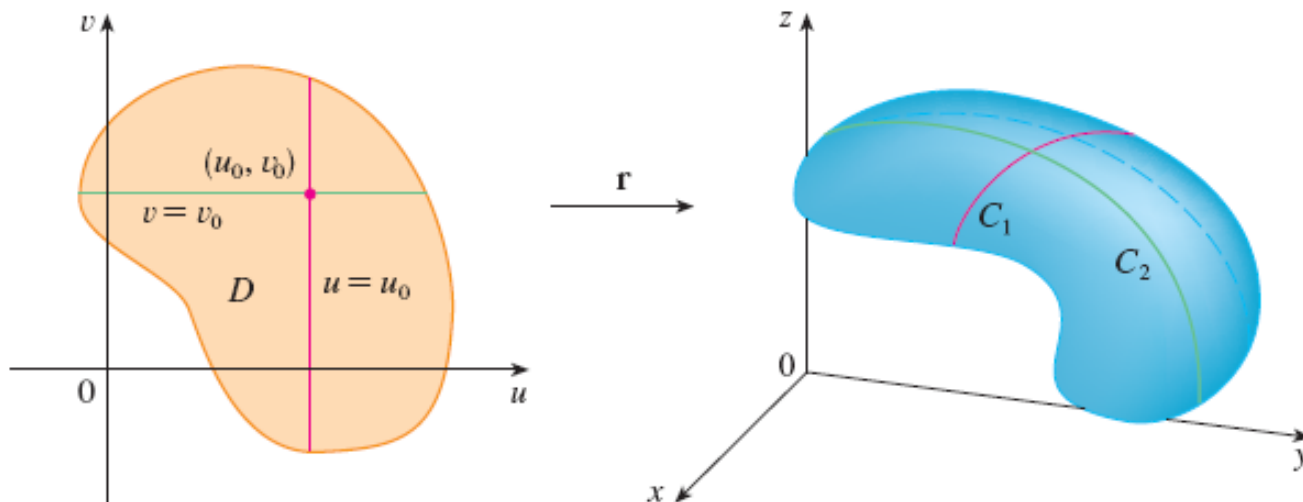
In much the same way that we describe a space curve by a vector function $\mathbf{r}(t)$ of a single parameter t , we can describe a surface by a vector function $\mathbf{r}(u, v)$ of two parameters u and v . We suppose that

$$\boxed{1} \quad \mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

is a vector-valued function defined on a region D in the uv -plane. So x , y , and z , the component functions of $\mathbf{r}$, are functions of the two variables u and v with domain D . The set of all points (x, y, z) in $\mathbb{R}^3$ such that

$$\boxed{2} \quad x = x(u, v) \quad y = y(u, v) \quad z = z(u, v)$$

and (u, v) varies throughout D , is called a **parametric surface** S and Equations 2 are called **parametric equations** of S . Each choice of u and v gives a point on S ; by making all choices, we get all of S . In other words, the surface S is traced out by the tip of the position vector $\mathbf{r}(u, v)$ as (u, v) moves throughout the region D . (See Figure 1.)



GRID CURVES

Example

V EXAMPLE 4 Find a parametric representation of the sphere

$$x^2 + y^2 + z^2 = a^2$$

EXAMPLE 5 Find a parametric representation for the cylinder

$$x^2 + y^2 = 4 \quad 0 \leq z \leq 1$$

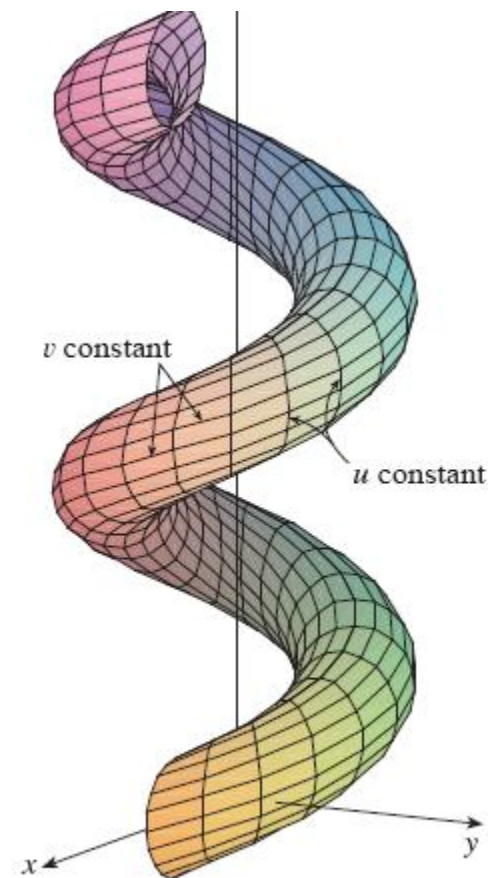
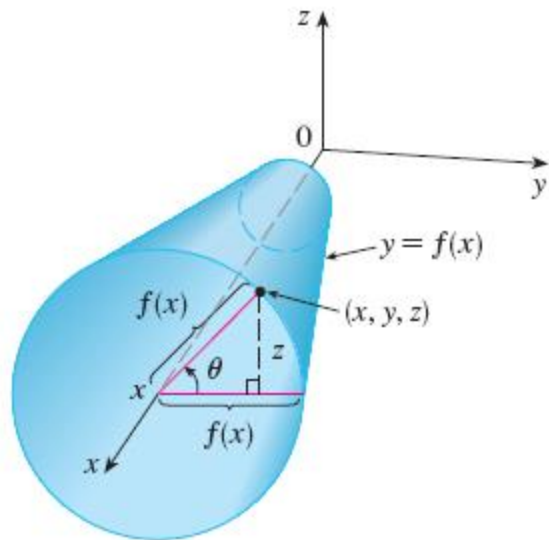


FIGURE 5

V EXAMPLE 6 Find a vector function that represents the elliptic paraboloid $z = x^2 + 2y^2$.

Surfaces of Revolution



3

$$x = x \quad y = f(x) \cos \theta \quad z = f(x) \sin \theta$$

FIGURE 10

EXAMPLE 8 Find parametric equations for the surface generated by rotating the curve $y = \sin x$, $0 \leq x \leq 2\pi$, about the x -axis. Use these equations to graph the surface of revolution.

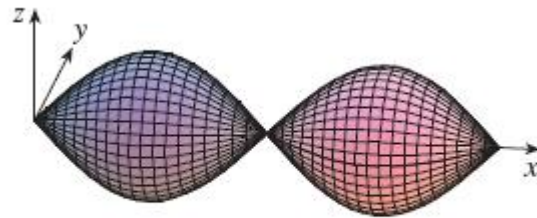
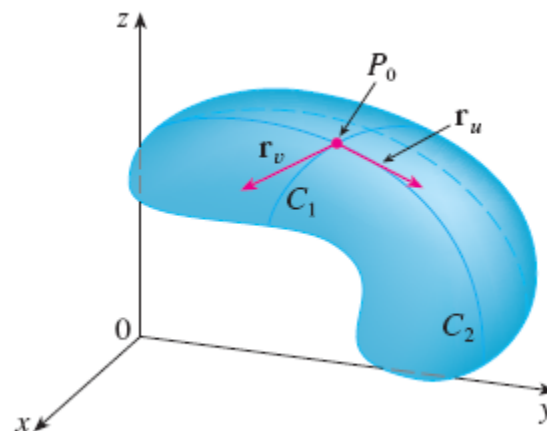
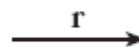
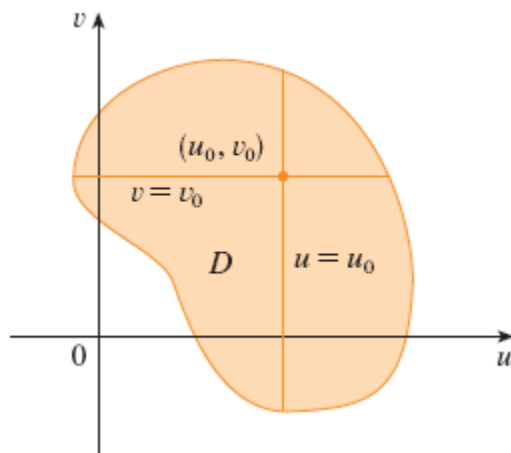
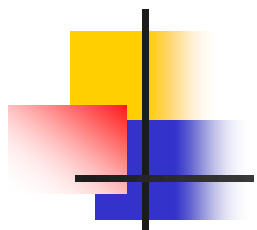


FIGURE 11

Tangent plane



4

$$\mathbf{r}_v = \frac{\partial X}{\partial v}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0)\mathbf{k}$$

5

$$\mathbf{r}_u = \frac{\partial X}{\partial u}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0)\mathbf{k}$$

If $\mathbf{r}_u \times \mathbf{r}_v$ is not $\mathbf{0}$, then the surface S is called **smooth** (it has no “corners”). For a smooth surface, the **tangent plane** is the plane that contains the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, and the vector $\mathbf{r}_u \times \mathbf{r}_v$ is a normal vector to the tangent plane.

V EXAMPLE 9 Find the tangent plane to the surface with parametric equations $x = u^2$, $y = v^2$, $z = u + 2v$ at the point $(1, 1, 3)$.

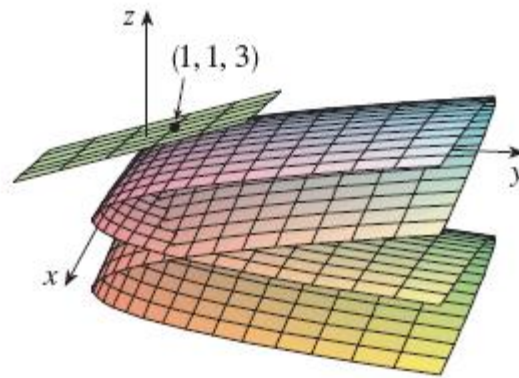


FIGURE 13

Back to change of variables

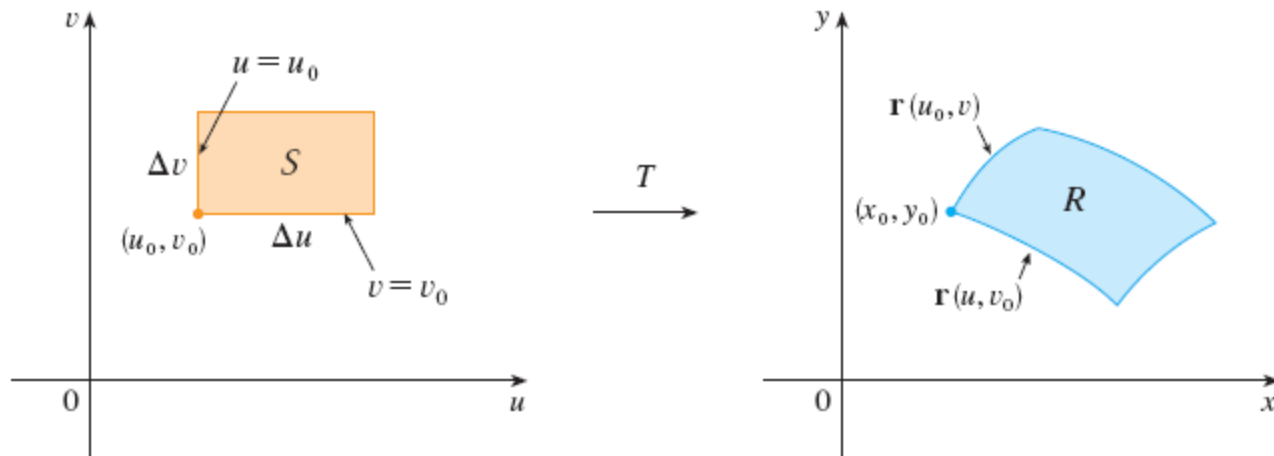


FIGURE 3

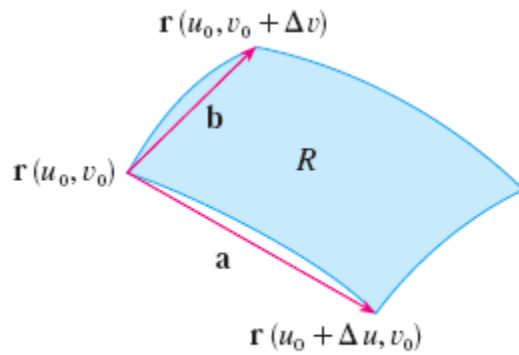
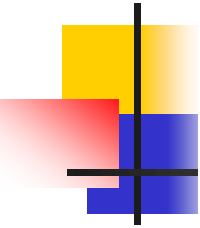


FIGURE 4

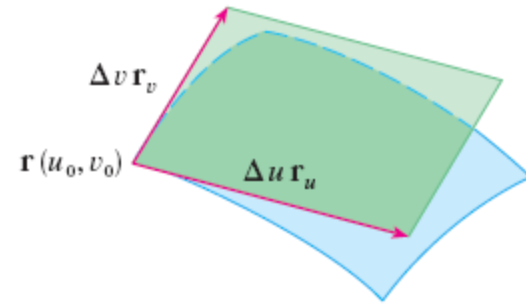
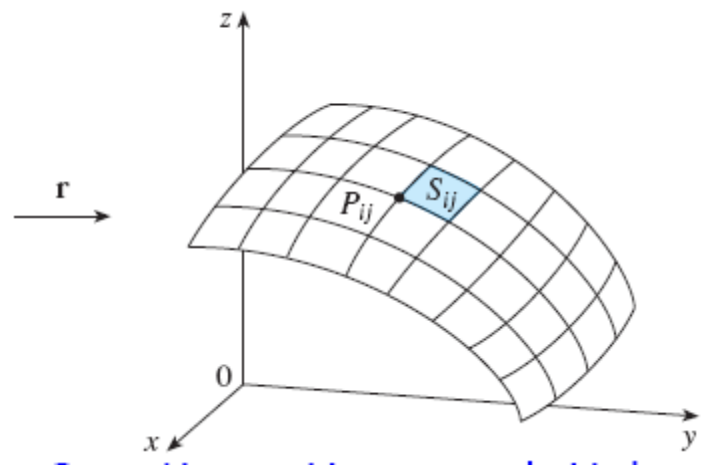
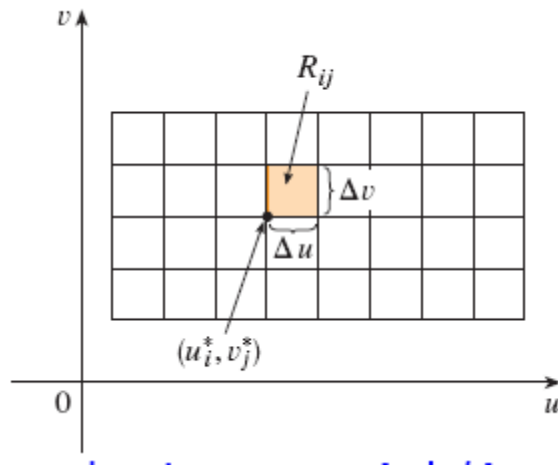


FIGURE 5

Surface Area



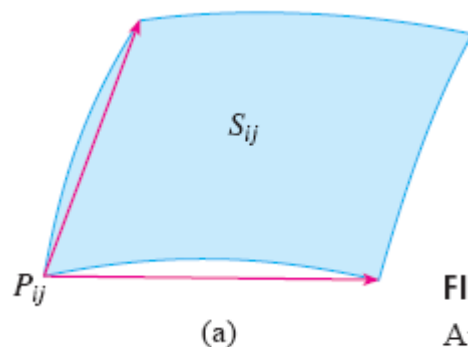
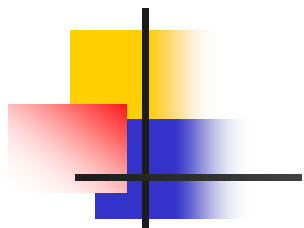
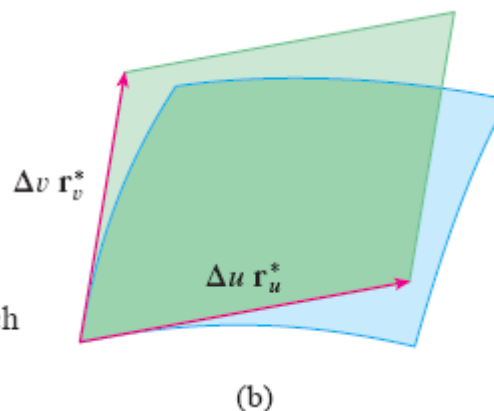


FIGURE 15

Approximating a patch
by a parallelogram



The area of this parallelogram is

$$|(\Delta u \mathbf{r}_u^*) \times (\Delta v \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

and so an approximation to the area of S is

$$\sum_{i=1}^m \sum_{j=1}^n |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

6 Definition If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D , then the **surface area** of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA$$

where

$$\mathbf{r}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k} \quad \mathbf{r}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}$$

EXAMPLE 10 Find the surface area of a sphere of radius a .

Surface Area of the graph of a function

For the special case of a surface S with equation $z = f(x, y)$, where (x, y) lies in D and f has continuous partial derivatives, we take x and y as parameters. The parametric equations are

$$x = x \quad y = y \quad z = f(x, y)$$

9

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

V EXAMPLE 11 Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$.

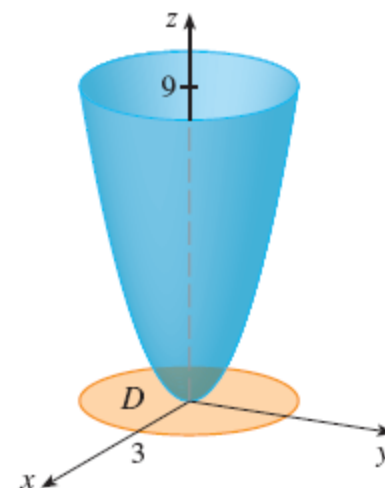
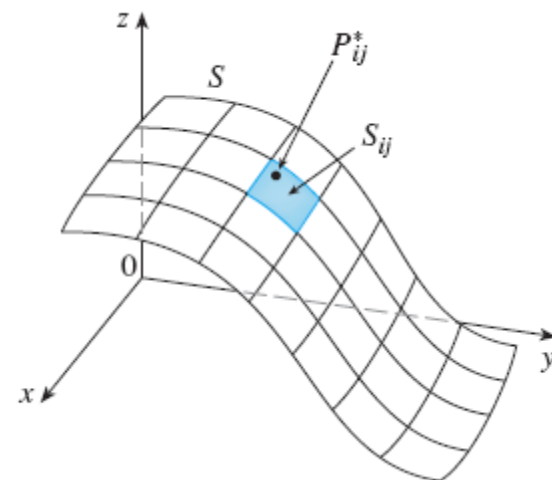
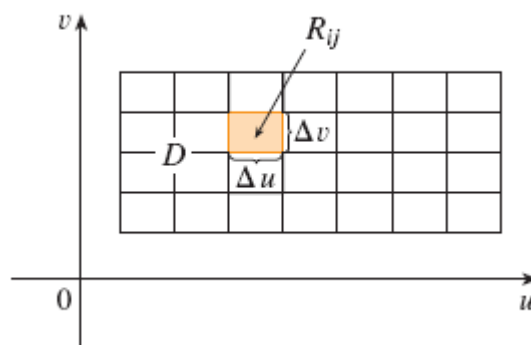
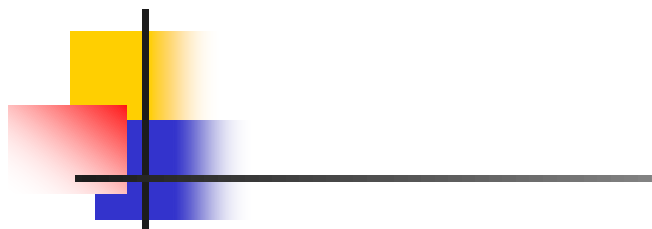


FIGURE 16

Surface Integrals



Parametric Surfaces

Suppose that a surface S has a vector equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

We first assume that the parameter domain D is a rectangle and we divide it into subrectangles R_{ij} with dimensions Δu and Δv . Then the surface S is divided into corresponding patches S_{ij} as in Figure 1. We evaluate f at a point P_{ij}^* in each patch, multiply by the area ΔS_{ij} of the patch, and form the Riemann sum

$$\sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \Delta S_{ij}$$

Then we take the limit as the number of patches increases and define the **surface integral of f over the surface S** as

1

$$\iint_S f(x, y, z) dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \Delta S_{ij}$$

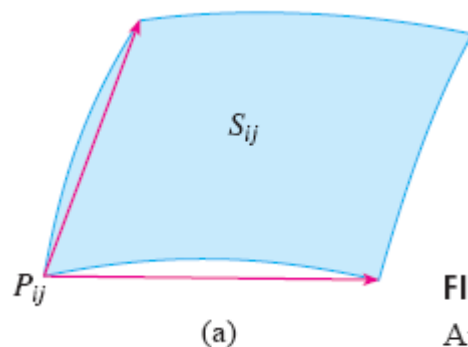
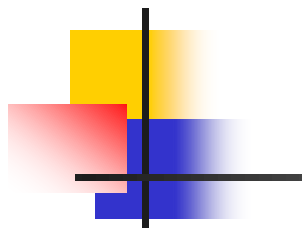
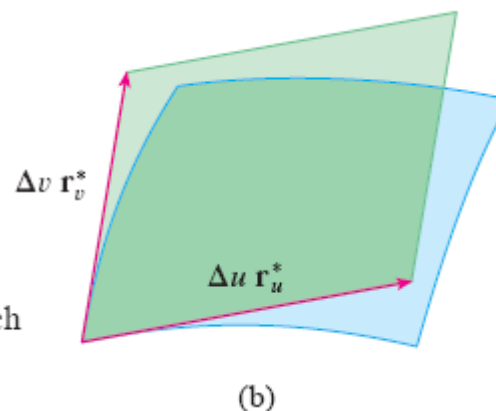


FIGURE 15

Approximating a patch by a parallelogram



To evaluate the surface integral in Equation 1 we approximate the patch area ΔS_{ij} by the area of an approximating parallelogram in the tangent plane. In our discussion of surface area in Section 16.6 we made the approximation

$$\Delta S_{ij} \approx |\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v$$

2

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

EXAMPLE 2 Compute the surface integral $\iint_S x^2 dS$, where S is the unit sphere $x^2 + y^2 + z^2 = 1$.

SURFACE INTEGRAL OF THE GRAPH OF A FUNCTION

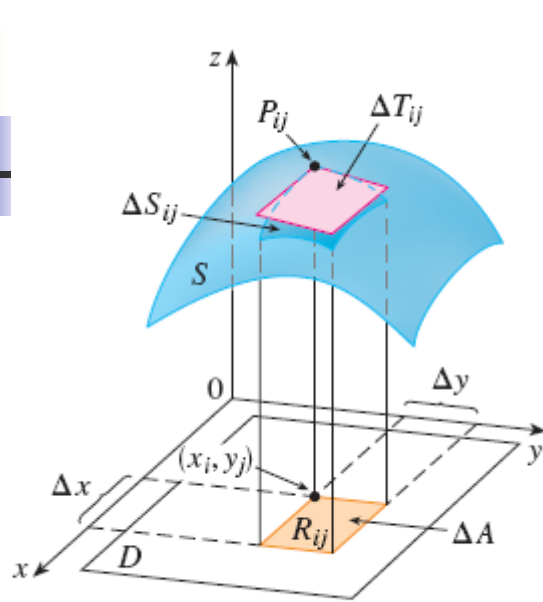


FIGURE 1

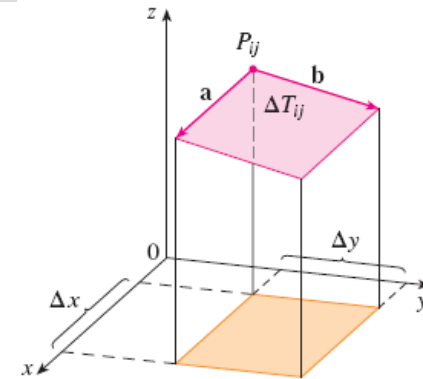


FIGURE 2

$$4 \quad \iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

EXAMPLE 2 Evaluate $\iint_S y \, dS$, where S is the surface $z = x + y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 2$.
(See Figure 2.)

$$\frac{13\sqrt{2}}{3}$$

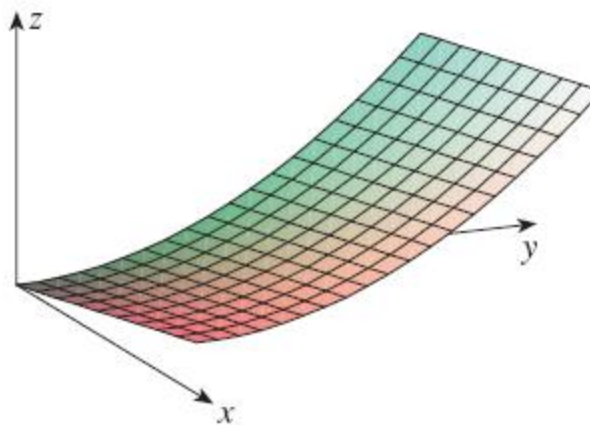


FIGURE 2

Mass & Center of Mass



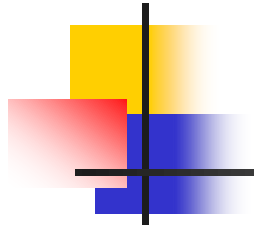
if a thin sheet (say, of aluminum foil) has the shape of a surface S and the density (mass per unit area) at the point (x, y, z) is $\rho(x, y, z)$, then the total mass of the sheet is

$$m = \iint_S \rho(x, y, z) dS$$

and the **center of mass** is $(\bar{x}, \bar{y}, \bar{z})$, where

$$\bar{x} = \frac{1}{m} \iint_S x \rho(x, y, z) dS \quad \bar{y} = \frac{1}{m} \iint_S y \rho(x, y, z) dS \quad \bar{z} = \frac{1}{m} \iint_S z \rho(x, y, z) dS$$

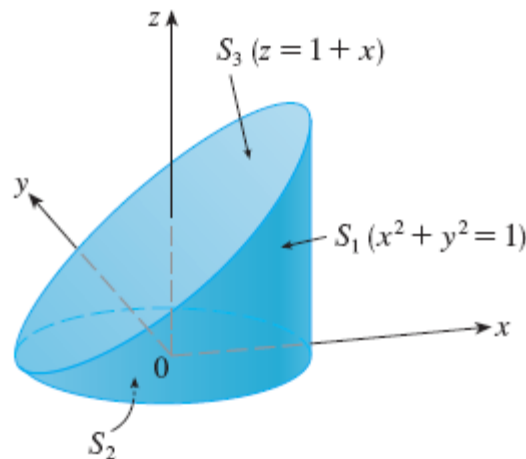
SURFACE INTEGRAL FOR PIECEWISE SMOOTH SURFACES



If S is a piecewise-smooth surface, that is, a finite union of smooth surfaces $S_1, S_2, \dots, S_n$ that intersect only along their boundaries, then the surface integral of f over S is defined by

$$\iint_S f(x, y, z) \, dS = \iint_{S_1} f(x, y, z) \, dS + \cdots + \iint_{S_n} f(x, y, z) \, dS$$

V EXAMPLE 3 Evaluate $\iint_S z \, dS$, where S is the surface whose sides S_1 are given by the cylinder $x^2 + y^2 = 1$, whose bottom S_2 is the disk $x^2 + y^2 \leq 1$ in the plane $z = 0$, and whose top S_3 is the part of the plane $z = 1 + x$ that lies above S_2 .



$$\left(\frac{3}{2} + \sqrt{2}\right)\pi$$

FIGURE 3